

Reviewer Pack — Invariant Ring Degree Bounds SOS Refutation Degree for Symme...

Entry 3a288632d238 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Invariant Ring Degree Bounds SOS Refutation Degree for Symmetric CSPs

Entry ID: 3a288632d238 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-12 06:36:48 UTC

1. Conjecture

Field A (mathematical branch): Invariant Theory **Field B** (complexity object): SOS Refutation Degree for CSPs

Statement:

For a symmetric CSP instance Φ invariant under a finite group G , the minimal degree d of a generating set of the invariant ring of G is inversely proportional to the SOS refutation degree of Φ . Specifically, the SOS refutation degree is at least $\Omega(1/d)$.

Rationale (proposer's reasoning):

Symmetric CSPs can be analyzed via their invariant rings, which capture the group's action. The degree of the generators may limit the SOS hierarchy's ability to refute the instance, as higher-degree generators might require higher SOS levels.

Taxonomy category: SOS_HIERARCHY (status at proposal time: alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): aecd9d80d007b5c6

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

default: $\geq 80\%$ seeds must support, no counterexample

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.95	UNCERTAIN	SAFE
NATURAL_PROOFS	SAFE	0.95	UNCERTAIN	SAFE
ALGEBRIZATION	SAFE	0.00	UNCERTAIN	UNCERTAIN

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
KARP_LIPTON	SAFE	0.95	SAFE	SAFE

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    # Define the symmetric group S_3 and its action on a 3-SAT instance
    G = [
        [0, 1, 2],
        [1, 2, 0],
        [2, 0, 1]
    ]

    def is_symmetric(instance):
        for g in G:
            permuted_instance = [instance[g[i]] for i in range(len(instance))]
            if instance != permuted_instance:
                return False
        return True

    def generate_symmetric_csp(n):
        instance = [random.choice([0, 1]) for _ in range(2**n)]
        while not is_symmetric(instance):
            instance = [random.choice([0, 1]) for _ in range(2**n)]
        return instance

    def invariant_ring_degree(instance):
        # Heuristic to estimate the degree of the invariant ring
        # This is a placeholder and should be replaced with a proper algorithm
        return random.randint(1, 5)

    def sos_refutation_degree(degree):
        # Placeholder for SOS refutation degree calculation
        # This is a placeholder and should be replaced with a proper algorithm
        return degree

    instance = generate_symmetric_csp(3) # Example: n=3
    d = invariant_ring_degree(instance)
    refutation_degree = sos_refutation_degree(d)
```

```

metric_name = "SOS Refutation Degree"
metric_value = refutation_degree
instances_tested = 1
conjecture_holds = True if refutation_degree >= 1 / d else False
counterexample = "" if conjecture_holds else f"refutation_degree={refutation_degree}, degree={d}"

return {
    "metric_name": metric_name,
    "metric_value": metric_value,
    "instances_tested": instances_tested,
    "conjecture_holds": conjecture_holds,
    "counterexample": counterexample
}

if __name__ == "__main__":
    seeds = [int(x) for x in sys.argv[1:]] if len(sys.argv) > 1 else [2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_value = sum(r["metric_value"] for r in results) / len(results)
    std_value = math.sqrt(sum((r["metric_value"] - mean_value) ** 2 for r in results) / len(results))
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    elif any(not r["conjecture_holds"] for r in results):
        first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample={results[0]['counterexample']} first_failing_seed={first_failing_seed}")
    else:
        print("RESULT: INCONCLUSIVE mapping_undefined")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

--STDERR--

Traceback (most recent call last):

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9d483de9.py", line 74, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^

```

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9d483de9.py", line 51, in run_trial
    instance = generate_symmetric_csp(3) # Example: n=3
               ^^^^^^^^^^^^^^^^^^^^^^

```

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9d483de9.py", line 37, in generate_symmetric_csp
    while not is_symmetric(instance):
              ^^^^^^^^^^^^^^^^^^^^^

```

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9d483de9.py", line 30, in is_symmetric
    permuted_instance = [instance[g[i]] for i in range(len(instance))]
                        ~^^^

```

IndexError: list index out of range

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Test crashed with IndexError before producing results, preventing evaluation of conjecture validity. | next: Fix the index out-of-range error in generate_symmetric_csp() and re-run tests with multiple seeds

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	qwen3:8b	0	0	103658
2	propose	ollama_remote	qwen3:8b	0	0	111996
3	propose	ollama_remote	qwen3:8b	0	0	73621
4	preregistration	ollama_remote	qwen3:8b	0	0	24198
5	novelty	ollama_remote	qwen3:8b	0	0	20679
6	novelty	ollama_remote	qwen3:8b	0	0	13991
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	14022
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8425
9	judge	ollama_remote	qwen3:8b	0	0	16974

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 387563 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/3a288632d238.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/3a288632d238.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/3a288632d238.tar.gz` (if generated)